The Challenge of Variation in Medical Practice

Brent C. James, MD, Mstat; M. Elizabeth H. Hammond, MD

• Medicine has been identified as a profession for almost 3000 years based on a core premise that physicians have the right to evaluate their own quality. Because medicine is a profession and because of the special privileges granted to physicians by society, quality-based principles that evolved in the manufacturing business have been difficult to adapt to medical practice. Physicians learn from other physicians and medical literature. This leads to wide variation in what is considered best practice. Variation has complex association, including the variation in expert opinion, the complexity of medical knowledge, the variation in physician decision-making potential, and human error. Guidelines or algorithms are a strategy that are finding favor as a solution. The control of variation through guideline development, iterative refinement of guidelines, and feedback to physicians will improve medical practice. By removing variation, physicians can honor the fiduciary trust that they have made to patients, make reasoned decisions, improve outcomes, and focus attention on making medical improvements.

(*Arch Pathol Lab Med.* 2000;124:1001–1003)

Medicine has been identified as a profession for almost 3000 years. This long history stems from a core premise: physicians reserve to themselves the right to evaluate their own quality. Certain sentinel principles underlie that assertion.

First, physicians bear a fiduciary trust in regard to their patients' health care needs—physicians agree to place their patients' health before any other outcome or goal and swear an oath to act as their patients' health care advocates.

Second, physicians maintain a special body of knowledge that they apply on behalf of their patients under that fiduciary trust. Medical knowledge is complex and difficult and is not generally available outside the profession. Because of physicians' specialized knowledge applied under a fiduciary trust, society grants special privileges to physicians—society allows physicians to perform functions that are considered criminal if done by other, unanointed hands. In addition to applying medical knowledge on behalf of their patients (as physician practitioners), physicians also agree to transmit their medical knowledge to a next generation of new, young physicians entering the profession (as physician teachers). Physicians also agree to improve the knowledge they transmit to future generations, beyond what they received themselves on entering the profession (as physician scientists).

Third, if the preceding 2 principles are correct, then the only person qualified to evaluate a physician's performance, on behalf of a patient, is another physician. Only another physician has the necessary knowledge and experience to judge whether a professional colleague adequately discharged his or her fiduciary trust to a particular patient. This is the reason that the profession of medicine can legitimately reserve the right to evaluate its own quality. This aspect of professional responsibility is far-reaching: not only do physicians resist attempts of those outside the profession to inappropriately judge medical performance, they also insist on holding one another accountable for their performance within the profession of medicine.

Because medicine is a profession and because of the special privileges granted to physicians by society, quality-based principles that evolved in the manufacturing business have been difficult to adapt to medical practice. Physicians learn from other physicians and the medical literature. This leads to wide variation in what is considered best practice. Physicians are granted significant autonomy in their decision making. They are not required to justify their decisions based on best universal practice; decisions must only be justified on the basis of what the physicians consider best for each individual patient. One of the major reasons for this situation is that it is expensive and time-consuming to determine best practice for all medical situations. Performing trials of one strategy versus another in a randomized controlled fashion takes significant time and money and is therefore not widely applied to all aspects of medicine. It is difficult to find similar groups of patients for the purpose of such investigations without resorting to national trials, hundreds of practitioners, and thousands of patients. Thus, variation in medicine is understandable but certainly not desirable. Recent work in many medical disciplines has drawn attention to wide variation in medical practices and the outcomes associated with those practices.

Variation in medicine has complex associations. Experts vary in their opinions, both in practice and in print, which confuses physicians in determining best practice.[1–3] The complexity of medical knowledge and the rapid ongoing evolution of that knowledge cause further confusion.[4–6] Furthermore, individual physicians vary in decision-making capacity from time to time based on health, time constraints, distractions, and level of stress. Human error is

Accepted for publication December 17, 1999.

From Intermountain Health Care (Dr James) and LDS Hospital (Dr Hammond), Salt Lake City, Utah.

Presented at the College of American Pathologists Conference XXXV: Solid Tumor Prognostic Factors: Which, How and So What?, Chicago, Ill, June 10–13, 1999.

Reprints: Brent C. James, MD, Intermountain Health Care, 36 S State St, 16th Floor, Salt Lake City, UT 84111 (e-mail: bjames@ihc.com).

thus a universal problem, regardless of dedication, intelligence, or effort applied.[7] Finally, the human mind is only able to consider 5 to 7 variables at a time when making a decision,[8] and almost no medical decisions can be made without consideration of many more variables than that. The principal strategy used to cope with these difficulties is reliance on expert clinical judgment developed by watching and learning from others. Many investigations have shown that such expert opinion is highly variable.[9,10] Guidelines or algorithms as a strategy to deal with complexity are only just beginning to find favor as a solution to the variation problem.

GUIDELINES AND THE CONTROL OF MEDICAL PRACTICE VARIATION

There are numerous examples of the power of controlling variation in improving the quality of medical practices and the outcomes of those practices.[11] Two examples from Intermountain Health Care serve to illustrate this concept.

To evaluate the efficacy of extracorporeal carbon dioxide removal as a therapeutic strategy to improve survival of patients with acute respiratory distress syndrome (ARDS), Dr Allen Morris and his colleagues at LDS Hospital (Intermountain Health Care's flagship hospital, located in Salt Lake City, Utah) reasoned that usual ventilator management must be rendered more uniform to serve as a valid control for this randomized, interventional study. He and his team developed an evidence-based, expert-consensus ventilator management protocol that the care team could apply 24 hours per day to all patients with ARDS. The protocol development process was difficult and prolonged. The final document included more than 40 pages of flowcharts with more than 20 decision nodes per page, totaling more than 840 specific recommendations regarding how to set a mechanical ventilator for a patient with ARDS based on physiologic presentation. On completing the protocol, Dr Morris realized that he had no validation data. There was no evidence that this protocol would necessarily produce ''best'' results. Dr Morris, therefore, trained intensivists at LDS Hospital in the use of the guideline and asked that they follow its recommendations, but he noted that the protocol had not been validated and that it would therefore probably need refinement during use. Clinicians interacting with the system could decide at each decision step whether to follow specific protocol therapy recommendations based on their clinical judgment. If a clinician decided not to use the protocol in any particular instance, the clinician recorded the reason for the deviation from the protocol. The entire team met for an hour each week to review each instance where a clinician deviated from protocol to (1) change the protocol, (2) agree that the protocol was correct, or (3) decide that the protocol deviation was justified but represented a rare event. When it was a rare event, the team should treat that deviation as random noise in the system (that is, not change practice but not add it to the protocol to keep the protocol of a more manageable, usable size). For the first patient with ARDS managed under the protocol, clinicians followed the protocol's recommendations only 41% of the time. After the team used the iterative review process for 8 patients, refining the document with more than 100 changes in 4 months, the protocol finally achieved more than 90% compliance. Use of the protocol resulted in significant improvements in clinical care. Specifically, physicians used less time to interact with their patients and did it in a more consistent manner. Patient care was more coordinated and streamlined, and there was less conflict between caregivers and confusion about the treatment plan. Patients left the intensive care unit more rapidly, because the care process was being managed 24 hours per day (rather than waiting for a physician to advance a patient's care by writing orders after morning or evening rounds). Total treatment costs decreased by about 25% per patient because of more efficient, standardized support processes and because of fewer errors arising from dropped handoffs among clinical team members. Best of all, patient survival significantly improved, increasing from 9.5% to 44%.[12]

Another example of the improvement of care through protocol development was the standardization of breast cancer reports at LDS Hospital. Dr Elizabeth Hammond, chairperson of the Department of Pathology in Intermountain Health Care's Urban Central Region, and her colleagues noticed that confusion regarding breast cancer pathology reports generated many telephone calls to the department from practicing physicians. Using expert opinion from the medical literature and discussion with oncologists about what information should appear in a breast cancer pathology report, Dr Hammond and her colleagues developed a list of ''synoptic elements''—those critical elements that should be included in every breast cancer pathology report. After evaluation by pathologists, the report format was refined and implemented widely throughout all Urban Central Region hospitals. This intervention resulted in a decrease in the amount of information missing from breast cancer pathology reports, which led to a proportional decrease in the number of telephone calls from physicians to the department. A satisfaction survey of oncologists after this strategy was implemented indicated that 100% of them felt that the synoptic reports were an improvement over the previous reports and resulted in facilitation of patient care.[13] The strategy proved so successful that it was implemented for 40 tumor types. A subsequent review of breast cancer reports 3 years after implementation confirmed that the reports were consistently provided in the new format without missing or confusing information.

In the cited examples and others, there is a common pattern that results in successful implementation of protocols to reduce medical practice variation: (1) measure practice variation within a well-defined clinical process using either direct statistical comparisons among physicians (which, unfortunately, requires very large numbers of patients) or a practice protocol to measure variations from a common baseline (possible with much smaller patient groups but requires a more sophisticated understanding of the proper role of protocols and guidelines as measurement baselines as opposed to practice mandates); (2) feed the variance data back to a professional peer group so that physicians and nurses can discuss best patient care with others who see similar patients in the same setting and supplement that discussion with literature reviews (evidence-based medicine), expert consultation, and, occasionally, special studies (This is a form of academic detailing, one of the only methods ever consistently proven to change physician practice.[14]); (3) based on the group's response, iteratively refine the protocol, improve the data system, and modify clinical practice; also standardize clinical infrastructure processes to provide consistent, predictable support for implementation so that physicians can trust that the process will be correctly followed unless there is a specific need to modify that process for a specific patient's unique need.

SUMMARY

Medical processes are extremely complex. Valid clinical knowledge about appropriate responses to this complexity is limited.[4–6] In many cases, evidence is poor or fragmentary. In addition, subjective judgment is notoriously poor across groups over time and is subject to individual experience. Finally, the problem of human error is a consistent difficulty that is experienced by physicians and other professionals over time. Control of variation is a sensible approach to reducing complexity and improving outcome. Without consistent behavior, outcomes cannot even be measured, let alone systematically improved. Writing a protocol or guideline is only the first step and may be the easiest one. To be useful, protocols require follow-up and feedback. A good guideline brings individuals together to collaborate on a common approach so that variance data can be generated that will change and improve practice. Using the process of feedback and refinement, even a mediocre protocol can evolve into demonstrated best care for patients.

By removing variation in our medical practice one process of care at a time, we can truly honor the fiduciary trust that we owe our patients when they place their lives in our hands. We can make reasoned decisions based on scientific evidence and refined by experience. By protecting our clinical performance from known sources of human error, including those arising from unnecessary complexity, by standardizing clinical support processes, we can improve our own medical decisions and the outcomes we achieve for our patients. Such standardization not only saves money, it also saves time, allowing physicians to focus on those areas where an individual patient needs a knowledgeable professional to modify the common protocol to his or her particular circumstance—the more challenging and intellectually rewarding parts of a patient's care. In these ways, we have the ability to craft a new profession for the new millennium, where trust is honored and physicians have more time and opportunity to chart new paths into unknown medical territory.

References

1. Mulley AG, Eagle KA. What is inappropriate care? *JAMA.* 1988;260:540–541.
2. Park RE, Fink A, Brook RH, et al. *Physician Ratings of Appropriate Indications for Six Medical and Surgical Procedures.* Santa Monica, Calif: Rand Corp; July 1986.
3. Park RE, Fink A, Brook RH, et al. Physician ratings of appropriate indications for six medical and surgical procedures: theoretical indications vs. indications used in practice. *Am J Public Health.* 1989;79:445–447.
4. Williamson JW, Goldschmidt PG, Jillson IA. *Medical Practice Information Demonstration Project: Final Report.* Baltimore, Md: Policy Research Inc; 1979. Office of the Assistant Secretary of Health, Department of Health, Education, and Welfare, contract 28277-0068GS.
5. Institute of Medicine. *Assessing Medical Technologies.* Washington, DC: National Academy Press; 1985:5.
6. Ferguson JH. Research on the delivery of medical care using hospital firms: proceedings of a workshop, April 30 and May 1, 1990, Bethesda, Maryland. *Med Care.* 1991;29(suppl 7):JS1–JS2.
7. McDonald CJ. Protocol-based computer reminders, the quality of care and the non-perfectibility of man. *N Engl J Med.* 1976;295:1352–1355.
8. Nicolis JS, Tsuda I. Chaotic dynamics of information processing: the "magic number seven plus-minus two" revisited. *Bull Math Biol.* 1985;47:343–365.
9. Eddy D. *A Manual for Assessing Health Practices & Designing Practice Policies: The Explicit Approach.* Philadelphia, Pa: American College of Physicians; 1992:14.
10. Eddy D. Variations in physician practice: the role of uncertainty. *Health Aff (Millwood).* 1984;3:74.
11. James BC. TQM in clinical medicine. *Front Health Serv Manage.* 1991; 7(4):44–48.
12. Morris AH. Computerized protocols and bedside decision support. *Crit Care Clin.* 1999;15:523–545.
13. Hammond EH, Flinner RL. Clinically relevant breast cancer reporting: using process measures to improve anatomic pathology reporting. *Arch Pathol Lab Med.* 1997;121:1171–1175.
14. Soumerai SB, Avorn J. Principles of educational outreach to improve clinical decision making. *JAMA.* 1990;263:549–556.